

AMENDMENT NO. 2 MARCH- 1996
TO
IS 8446 : 1991 PESTICIDE — CARBENDAZIM
(MBC) WP — SPECIFICATION

(First Revision)

(Page 2, clause 7.3) — Substitute 'IS 1070 :1992' for 'IS 1070 : 1977'.

(Page 2, Annex A) — Substitute 'IS 1070 : 1992 Reagent grade water (third revision)' for '1070 : 1977 Water for general laboratory use (second revision)'.

(Page 2, Annex B) — Substitute the following for the existing:

‘ANNEX B

DETERMINATION OF CARBENDAZIM **(MBC)** CONTENT
[Table 1, *Item* (i)]

B-1 GENERAL

Either of the two methods, namely, High Performance Liquid Chromatography **(HPLC)** method (see B-2) or titration method (see B-3) shall be **used** for the determination of carbendazim content. However, HPLC method shall be the referee method.

B-2 BPLC METHOD

B-2.1 Principle

The sample is dissolved in methanol and **sulphuric** acid along with preliminary solvent mixture and the active ingredient content is determined at 281 **nm** by comparing the peak areas of carbendazim standard and sample.

B-2.2 Apparatus

High Performance Liquid Chromatograph — Equipped with a **printer-plotter-cum-integrator** and ultra-violet **(UV)** detector. The suggestive operating conditions are given below and may be varied depending upon available **instrument** provided standardization is done:

Column — Stainless steel, C 18, 250 to 300 mm length and 4.0 to 4.6 mm ID.

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Mobile Phase — Mixture of distilled water, methanol and 0.5 M (1 N) sulphuric acid in **glass** distilled **water**, in the ratio of **65:35:0.5** (v/v/v).

Detector — **UV at 281 nm.**

Flow Rate — 1 ml/minute (approx).

B-2.2.1 *Microbe Syringe* — **20 µl.**

B-2.2.2 *Volumetric Flask* — 100 ml.

B-2.2.3 *Pipette* — 10 ml.

B-2.3 **Reagents**

B-2.3.1 *Sulphuric Acid* — 1 N.

B-2.3.2 1,4 — dioxane -- HPLC Grade.

B-2.3.3 Methanol — HPLC Grade.

B-2.3.4 *Whatman Filter Paper No. 42*

B-2.33 Glass Distilled Water

B-2.3.6 Preliminary Solvent Mixture — A mixture of **60** parts of **1,4-dioxane** and 40 parts (v/v) of 1 N **Sulphuric** acid.

B-2.3.7 *Carbendazim Reference Standard* — of known purity.

B-2.4 Procedure

B-2.4.1 *Preparation of Reference Standard Solution*

Weigh 0.5 g of carbendazim reference standard in a **100-ml** volumetric **flask** and make up the volume using the **preliminary** solvent mixture (see B-2.3.6). **Shake** the volumetric flask vigorously to ensure complete dissolution. Take out 10 ml of this solution with a pipette into another **100-ml** volumetric flask **and make up the** volume using the mobile phase.

B-2.4.2 *Preparation of Sample Solution*

Weigh accurately a quantity of sample equivalent to 0.5 g of active ingredient in a **100-ml** volumetric flask and make up the volume using the preliminary solvent mixture (see B-2.3.6). Shake the volumetric flask to ensure complete dissolution. Filter the solution through **Whatman Filter** Paper No. 42 or equivalent. Take out 10 **ml** of the filtrate into another **100-ml** volumetric flask and make up the volume using the mobile phase.

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B-2.4.3 Estimation

B-2.4.3.1 Inject 20 μ l of the reference standard solution until the peak area of 2 successive injections agree to within 2 **percent**. Subsequently, inject the standard and sample solutions in succession according to the following sequence:

R S₁ S₁ R S₂ S₂ R S₃ S₃ R

where

R = reference standard solution injections, and

S₁, S₂, S₃ = different sample solution injections.

B-2.4.3.2 Calculation

$$\text{Carbendazim content} = \frac{\text{Peak area of sample} \times \text{Mass of reference standard} \times \text{Percent purity of reference standard}}{\text{Peak area of reference standard} \times \text{Mass of sample}}$$

NOTES

1 It is advisable to use a Guard Column in **series** with the main column, of **the same** stationary phase, which **increases** the life of the main column.

2 Determine the **carbendazim** content by taking average of 2 injections.

H-3 TITRATION METHOD

H-3.1 Principle

Carbendazim is a weak base due to the tertiary nitrogen atom in the benzimidazole ring and can be titrated with **perchloric** acid in acetic acid in non-aqueous medium.

B-3.2 Reagents

B-3.2.1 **Acetic Acid** — glacial, anhydrous.

B-3.2.2 **Acetic Anhydride**

B-3.2.3 **Crystal Violet Indicator** — 0.5 percent (m/v) in acetic acid.

H-3.2.4 **Perchloric Acid** — 0.1 N, freshly prepared by dissolving 8.5 ml of 70 **percent perchloric** acid in glacial acetic acid. Add 15 ml of acetic anhydride cautiously, make up to 1 **litre** and keep overnight.

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B-3.3 Procedure

B-3.3.1 Weigh 1 **g** of the sample and transfer into a **60-ml** centrifuge tube. Add 25 **ml** of distilled water. Stir well. Rinse the glass rod with 15 ml of distilled water. Centrifuge the suspension at 2 500 RPM for **30** minutes. Decant the supernatant liquid without disturbing the sediment. Transfer the sediment quantitatively with **25** ml of glacial acetic acid into a clean dry conical flask. Rinse the centrifuge tube twice with 15 **ml** of glacial acetic acid. Add 10 ml of acetic **anhydride** to a **conical** flask Warm the contents at **50-70°C** on a water bath for 5 minutes. Cool to room temperature. Filter the contents under suction through a sintered glass crucible into a dry flask. Rinse the crucible thrice with **10-ml** of glacial acetic acid. **Discard** the residue and titrate the filtrate against 0.1 N perchloric-acid using crystal violet indicator. The end point is **from** blue to green.

B-3.3.2 Carry out a blank determination using the method given under B-3.3.1. •

B-3.3.3 Calculation

$$\frac{\text{Carbendazim (MBC) content, percent by mass}}{100} = \frac{19.1 \times N \times (V_1 - V_2)}{M}$$

where

N = normality of perchloric acid;

V_1 = volume, in ml, of standard perchloric acid required for sample;

V_2 = volume, in ml, of standard perchloric acid required for the blank determination; and

M = mass, in g, of the sample.

B-3.4 Standardization of **Perchloric** Acid

B-3.4.1 Procedure

B-3.4.1.1 Weigh 0.2 **g** of potassium hydrogen phthalate and dissolve in 50 ml glacial acetic acid by warming for a short time. Titrate the solutions with 0.1 N **perchloric** acid with two drops of crystal violet as indicator, the end point being blue to green.

B-3.4.1.2 **Carry** out blank determination with 50 ml of glacial acetic acid and two drops of crystal violet.

B-3.4.13 Calculation

$$\text{Normality} = \frac{4.898 \times M}{V_1 - V_2}$$

where

M = mass, in g, of potassium hydrogen pbthalate;

***V*₁** = volume, in ml, of perchloric acid required for potassium hydrogen pbthalate (see B-3.4.1.1); and

***V*₂** = volume in ml of perchloric acid required for blank determination (see B-3.4.1.2).'

(Page 3, Annex C, Title) — Delete '**AFTER ACCELERATED STORAGE**'.

(Page 3, clause C-1.1) — Delete and renumber the subsequent clauses.